Let G be a Lie group, $\psi: G \longrightarrow \mathrm{Symp}(M,\omega)$ a symplectic action. For $g \in G$, let

$$\mathrm{Ad}_g : \mathfrak{g} \longrightarrow \mathfrak{g}$$

denote the derivative at the identity of the map $a \longmapsto gag^{-1}$. Then

$$\mathrm{Ad} : G \longrightarrow GL(\mathfrak{g})$$

is the adjoint representation. We define the coadjoint representation

$$\mathrm{Ad}^* : G \longrightarrow GL(\mathfrak{g}^*), \quad g \longmapsto \mathrm{Ad}^*_g$$

by

$$\mathrm{Ad}^*_g(\xi) : X \longmapsto \xi(\mathrm{Ad}_{g^{-1}}(X)).$$

Note that

$$\{\text{sympl. actions } \mathbb{R} \circlearrowright M\} \longleftrightarrow \{\text{complete sympl. vector fields on } M\}$$

$$\psi \longmapsto \frac{\partial}{\partial t}\Big|_{t=0} \psi_t, \quad X \longmapsto \mathrm{flow}(X) = \exp(tX)$$

at P, $\frac{\partial}{\partial t}\big|_{t=0} \psi_t \cdot P$, ...

Let $G \circlearrowright M$, $X \in \mathfrak{g}$. Let $X^\#$ denote the vector field on M generated by $\{\exp tX\}$.

Def: $G \circlearrowright M$ is Hamiltonian if there is a moment map $\mu : M \longrightarrow \mathfrak{g}^*$ such that $\forall X \in \mathfrak{g}$,

"simultaneous Hamiltonian function"

$$d\mu^X = i_{X^\#}\omega,$$

where μ^X is the composition $M \xrightarrow{\mu} \mathfrak{g}^* \xrightarrow{X} \mathbb{R}$, and such that

$$\mu \circ \psi_g = \mathrm{Ad}^*_g \circ \mu \quad \forall g \in G.$$

$$\begin{array}{ccc} M & \xrightarrow{\psi_g} & M \\ {\scriptstyle\mu}\downarrow & & \downarrow{\scriptstyle\mu} \\ \mathfrak{g}^* & \xrightarrow{\mathrm{Ad}^*_g} & \mathfrak{g}^* \end{array}$$

Thm (Marsden–Weinstein–Meyer): Let $G \circlearrowright M$ be Hamiltonian with moment map $\mu : M \longrightarrow \mathfrak{g}^*$. Assume G is compact and $G \circlearrowright \mu^{-1}(0)$ freely. Then

- $M_{red} := \mu^{-1}(0)/G$ is a manifold;
- $\pi : \mu^{-1}(0) \longrightarrow M_{red}$ is a principal G-bundle;
- M_{red} has a symplectic form ω_{red} with $\omega|_{\mu^{-1}(0)} = \pi^*\omega_{red}$.

Thm (Noether): Let $G \circlearrowright M$ be Hamiltonian. Then $f : M \longrightarrow \mathbb{R}$ is G-invariant iff μ is constant along the integral curves of X_f.

Proof: Let γ be an integral curve of X_f. Then for any $X \in \mathfrak{g}$,

$$\begin{aligned} \frac{\partial}{\partial t}\mu^X(\gamma) &= d\mu^X(\dot\gamma) \\ &= (i_{X^\#}\omega)(X_f) \qquad \dot\gamma = X_f(\gamma),\ \mu \text{ moment map} \\ &= -(i_{X_f}\omega)(X^\#) \\ &= -df(X^\#). \end{aligned}$$

The last expression is zero for any $X \in \mathfrak{g}$ iff f is G-invariant. □

Let $\mathcal{X}^{sympl}$, $\mathcal{X}^{Ham}$ denote the space of symplectic, Hamiltonian vector fields on M. We get the following s.e.s. of Lie algebras

$$0 \longrightarrow \underset{\text{Lie bracket}}{\mathcal{X}^{Ham}} \longrightarrow \underset{\text{Lie bracket}}{\mathcal{X}^{sympl}} \xrightarrow{v \mapsto i_v\omega} \overset{\text{trivial bracket}}{H^1(M;\mathbb{R})} \longrightarrow 0$$

$$0 \longrightarrow \underset{\text{trivial bracket}}{\mathbb{R}} \longrightarrow \underset{\text{Poisson bracket}}{C^\infty(M)} \xrightarrow[(=df^\# \text{ via } \omega)]{f \mapsto X_f} \underset{\text{Lie bracket}}{\mathcal{X}^{Ham}} \longrightarrow 0$$

Recall that an action $\psi : G \longrightarrow \mathrm{Symp}(M,\omega)$ induces a map

$$d\psi : \mathfrak{g} \longrightarrow \mathcal{X}^{sympl}, \quad X \longmapsto X^\# = \text{v.f. generated by } \{\exp tX\}$$

If G is connected then the existence of a moment map $\mu : M \longrightarrow \mathfrak{g}^*$ is equivalent to the existence of a comoment map $\mu^* : \mathfrak{g} \longrightarrow C^\infty(M)$ such that

$$\begin{array}{ccc} \mathfrak{g} & \xrightarrow{d\psi} & \mathcal{X}^{sympl} \\ & {\scriptstyle\mu^*}\searrow \ \circlearrowleft & \nearrow{\scriptstyle f \mapsto X_f} \\ & C^\infty(M) & \end{array}$$

$\mu^* : X \longmapsto \mu^X = X \circ \mu$

$\mu : M \times \mathfrak{g} \longrightarrow \mathbb{R}$

Note that the moment map μ can then be considered as the restriction of the map dual to μ^* to M.

Prop: The equivariance of $\mu : M \longrightarrow \mathfrak{g}^*$ is equivalent to $\mu^* : \mathfrak{g} \longrightarrow C^\infty(M)$ being a Lie algebra anti-homomorphism.

Proof: We have

$$\begin{aligned} \{\mu^*(X), \mu^*(Y)\}(m) &= \{\mu^X, \mu^Y\}(m) && \text{def. of } \mu^* \\ &= \omega(X_{\mu^X}, X_{\mu^Y})(m) && \text{def. of } \{\cdot,\cdot\} \\ &= d\mu^X(X_{\mu^Y})(m) && \text{def. of } X_\cdot \\ &= X_{\mu^Y}(\mu^X)(m) \\ &= \frac{\partial}{\partial t}\mu^X(\exp tY \cdot m) && \text{since } X_{\mu^Y} = Y^\# \text{ (moment map)} \\ &= \frac{\partial}{\partial t} X(\mathrm{Ad}^*_{\exp tY}(\mu(m))) && \text{EQUIVARIANCE} \\ &= \frac{\partial}{\partial t}\mu(m)(\mathrm{Ad}_{\exp(-tY)}X) \\ &= -\mu^{[X,Y]}(m). \end{aligned}$$

□